Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q02] | Part A (Q1-Q5)

Answer: 021

1. The reflected points are $(v,u),(-v,u),(-v,-u),(v,-u)$. The rectangle $BCDE$ contributes $4uv$ and triangle ABE contributes $u(u-v)$.

2. Thus $u(u+3v)=451=11\cdot 41$. Because $u<u+3v<4u$, only $u=11, u+3v=41$ fits. Hence $v=10$ and $u+v=21$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2001-II-Q04] | Part A (Q1-Q5)

Answer: 067

1. Write $P=(8t,15t), Q=(10u,3u)$. The midpoint gives $8t+10u=16, 15t+3u=12$, so $t=4/7, u=8/7$.

2. The displacement is $(48/7,-36/7)$, whose length is $60/7$. Thus the answer is 67.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-I-Q04] | Part A (Q1-Q5)

Answer: 177

1. Use affine coordinates $A=(0,0), B=(1,0), D=(0,1), C=(1,1)$. Then MN has intercept equation $1000x/17+2009y/17=1$.

2. On diagonal $x=y=t$, this gives $3009t=17$. Thus $AC/AP=1/t=3009/17=177$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-II-Q03] | Part A (Q1-Q5)

Answer: 141

1. With $A = (0,0), B = (100,0), D = (0,h)$, vectors $AC = (100,h)$ and $BE = (-100,h/2)$ have dot product $-10000 + h^2/2$.

2. Perpendicularity yields $h = 100\sqrt{2}$, between 141 and 142. The answer is 141.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q02] | Part A (Q1-Q5)

Answer: 036

1. Put $D=(0,0), B=(12,10)$. For a unit vector (u,v) with positive components, the conditions give $E=(12-9u, 10-9v), F=(8u,8v)$.

2. Equal heights imply $17v=10$. Thus $u=\sqrt{1-100/289}=3\sqrt{21}/17$.

3. The horizontal separation is $|12-17u|=3\sqrt{21}-12$. The points are inside the rectangle and the line reaches AD, so the valid value gives $3+21+12=36$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q03] | Part A (Q1-Q5)

Answer: 031

1. The axes have equations $L : 5x - 12y - 132 = 0$ and $M : 12x + 5y - 90 = 0$. New coordinates are signed perpendicular distances to the opposite axes.

2. At the point, the M expression is -123, while it is positive at A, so the new horizontal coordinate is $-123/13$. The L expression is -526, with the same negative sign as at B, so the new vertical coordinate is $526/13$.

3. The sum is $403/13 = 31$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-II-Q02] | Part A (Q1-Q5)

Answer: 810

1. Take $B=(0,0), C=(s,0), A=(0,s), D=(s,s)$ and $E=(a,s), F=(b,0)$. Equal BE, FD gives $a=s-b$.

2. Then $EF=BE$ yields $(2a-s)^2=a^2$, so $a=s/3$ or $a=s$. The latter identifies $E=D, F=B$, excluded by the separately named nonvertex points.

3. Therefore $900=s^2+(s/3)^2=10s^2/9$, and the area is 810.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-II-Q04] | Part A (Q1-Q5)

Answer: 040

1. The midpoint is $(3/2, \sqrt{3})$. Add or subtract $\sqrt{3}/2$ times a ninety-degree rotation of $B - A$ to get the possible third vertices. The first-quadrant one is $C = (9/2, \sqrt{3}/2)$.

2. The centroid, also the center, is $(5/2, 5\sqrt{3}/6)$. Thus $xy = 25\sqrt{3}/12$, giving 40.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-I-Q04] | Part A (Q1-Q5)

Answer: 507

1. Choose $B=(0,0), A=(-16,0), C=(4,0)$. Equilateral apices are $D=(-8,8\sqrt{3}), E=(2,2\sqrt{3})$, so $M=(-7,\sqrt{3}), N=(-2,4\sqrt{3})$.

2. Use the determinant area directly: $x=\frac{1}{2}|(-7)(4\sqrt{3})-(-2)(\sqrt{3})|=13\sqrt{3}$. Thus $x^2=507$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-II-Q04] | Part A (Q1-Q5)

Answer: 023

1. Use coordinates $C=(0,0), A=(1,1), R=(2,1), O=(3,0), L=(3,-1), I=(2,-2), N=(1,-2), E=(0,-1)$. They give the required alternating lengths and exterior turns of forty-five degrees.

2. Lines AI, RN meet CO at $Y=(4/3,0), Z=(5/3,0)$ and intersect at $W=(3/2,-1/2)$. The region consists of four congruent corner triangles like CAY and two central ones like YZW.

3. The former have area $(4/3)\cdot 1/2=2/3$; the latter $(1/3)(1/2)/2=1/12$. Hence total area $4(2/3)+2(1/12)=17/6$, giving 23.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q01] | Part A (Q1-Q5)

Answer: 059

1. Take $A=(0,0), B=(9,0)$. Distance equations give $C=(15,8)$ and reflection $D=(-6,8)$.

2. The overlap is the triangle with base AB and top $AC \cap BD$. Equating $y=8x/15$ and $y=8(9-x)/15$ gives height $12/5$.

3. Area $9(12/5)/2=54/5$, so the answer is 59.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-II-Q04] | Part A (Q1-Q5)

Answer: 108

1. The direction $(0, 12)$ becomes $(12, 0)$, so $m = 90$. A clockwise quarter-turn around (x, y) maps (p, q) to $(x + q - y, y - p + x)$.

2. The origin image gives $x - y = 24, x + y = 18$, hence $(x, y) = (21, -3)$. Substitution also gives both remaining images, so the requested sum is $90 + 21 - 3 = 108$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-I-Q02] | Part A (Q1-Q5)

Answer: 109

1. Set $A=(0,3), B=(0,0), C=(11,0), D=(11,3)$. The stated orientation and lengths give $F=(28/5,36/5)$ and $E=(27/5,-21/5)$. These satisfy the distance equations and $A+C=E+F$.

2. The second rectangle is bounded by $9 \leq 4x+3y \leq 44$ and $-12 \leq 3x-4y \leq 33$. Throughout $0 \leq x \leq 11, 0 \leq y \leq 3$, the latter pair already holds.

3. The overlap is therefore a parallelogram of height 3 and horizontal width $(44-9)/4 = 35/4$. Its area is $105/4$, giving 109.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q05] | Part A (Q1-Q5)

Answer: 106

1. Take the circle radius R and vertices $(R,0),(0,R),(-R,0),(0,-R)$, with $P=(R\cos t, R\sin t)$.

2. Distance formulas give $PA\,PC = 2R^2|\sin t|$ and $PB\,PD = 2R^2|\cos t|$. Squaring and adding yields $(2R^2)^2 = 56^2 + 90^2 = 11236$.

3. The square area is half the squared diagonal, namely $2R^2 = 106$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2024-I-Q05] | Part A (Q1-Q5)

Answer: 104

1. Use $D=(0,0), C=(107,0), A=(0,-16), E=(e,0), H=(e,17), G=(e+184,17)$.

2. The circle center is $(e+92,-8)$ by the perpendicular bisectors of AD and HG. Its distances to D, H give $(e+92)^2+64=92^2+625$.

3. Since $e>0$, $e+92=95$, so $e=3$ and $CE=107-3=104$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-II-Q01] | Part A (Q1-Q5)

Answer: 468

1. Use line coordinates $A = 0, C = 26, F = 73, D = 40, B = 18, E = 57$. If $G = (u, v)$, subtract $(u - 26)^2 + v^2 = 1600$ and $(u - 40)^2 + v^2 = 900$ to obtain $u = 58$.

2. Then $v^2 = 576$, so height is 24 and base $BE = 39$. Area $39 \cdot 24/2 = 468$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-II-Q03] | Part A (Q1-Q5)

Answer: 503

1. Take $A=(0,0), E=(20,0), B=(0,h)$. The angle conditions and nonconvex order give $C=(14,h-14)$ and $D=(20,h-8)$. Same-side strictness requires $h>14$.

2. Shoelace gives area $20h-164$. Divisibility by sixteen is $20h-164\equiv 0 \pmod{16}$, or $h\equiv 1 \pmod{4}$.

3. The possible integers are $17,21,\ldots,2025$, numbering $(2025-17)/4+1=503$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-II-Q08] | Part B (Q6-Q10)

Answer: 110

1. Use $B=(0,0), A=(\sqrt{11},0), C=(0,h), D=(d,h)$. The diagonal dot product gives $h^2=\sqrt{11}d$.

2. Put $u=h^2$. The leg condition is $(u-11)^2/11+u=1001$, yielding $u^2-11u-10890=0=(u-110)(u+99)$. Since $u>0$, $BC^2=110$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2004-II-Q07] | Part B (Q6-Q10)

Answer: 293

1. Folding preserves $EB' = EB = 17$, so right triangle AEB' gives $AB' = 15$, while $AB = 25$. Set $A = (0,0), B = (25,0), B' = (0,15), E = (8,0)$.

2. The crease is the perpendicular bisector of BB', whose slope is $-3/5$; its own equation is $y = 5(x-8)/3$. At F, $x = 25 - 3 = 22$, giving height $BC = 70/3$.

3. The perimeter is $2(25 + 70/3) = 290/3$, so the sum is 293.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-I-Q10] | Part B (Q6-Q10)

Answer: 047

1. The midpoint is $(35/2, 39/2)$, so the median line is $q = -5p + 107$.

2. Signed double area gives $|980 - 56p| = 140$, hence $p = 15$ or 20. Since $p + q = 107 - 4p$, its maximum is 47.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-I-Q10] | Part B (Q6-Q10)

Answer: 065

1. The line through $(1, 1/2)$ and $(3/2, 2)$ has slope three. Each point is the symmetry center of one pair of tangent disks, so the line bisects those four disks in pairs.

2. It misses the other four disks, leaving two on each side; their centers have signed line values in opposite pairs of safe magnitudes. Thus it bisects the union. Its equation is $y = 3x - 5/2$, or $6x = 2y + 5$.

3. The answer is $36 + 4 + 25 = 65$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-II-Q06] | Part B (Q6-Q10)

Answer: 012

1. Take $A=(0,0), B=(1,0), C=(1,1), D=(0,1)$. Equal lengths $AE=AF$ force $E=(1,t), F=(t,1)$. Then $1+t^2=2(1-t)^2$, giving $t=2-\sqrt{3}$.

2. For smaller side u, its opposite vertex is $(1-u,u)$ and lies on $y=tx$. Thus $u=t/(1+t)=(3-\sqrt{3})/6$, so the requested sum is 12.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2009-II-Q10] | Part B (Q6-Q10)

Answer: 096

1. Take $A=(0,0), B=(5,0), C=(5,12)$. The bisector through A has slope $(12/13)/(1+5/13)=2/3$.

2. At C, the ray toward D is the reflection of CA across vertical line CB, so it lies through $(10,0)$, with equation $5y=120-12x$.

3. Intersection gives $D=(180/23, 120/23)$, hence $AD=60\sqrt{13}/23$ and the answer 96.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2010-II-Q09] | Part B (Q6-Q10)

Answer: 011

1. Normalize the circumradius to 1, taking $A=(1,0), B=(1/2,\sqrt{3}/2), C=(-1/2,\sqrt{3}/2), F=(1/2,-\sqrt{3}/2)$. Then $H=(0,\sqrt{3}/2), G=(3/4,\sqrt{3}/4)$.

2. A small-hexagon vertex is $AH\cap FG$. Its equations $x+2y/\sqrt{3}=1$ and $y=3\sqrt{3}x-2\sqrt{3}$ give $(x,y)=(5/7,\sqrt{3}/7)$.

3. Sixty-degree symmetry makes the inner hexagon concentric and regular. The area ratio is the squared circumradius ratio, $25/49+3/49=4/7$. Hence $m+n=11$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-I-Q07] | Part B (Q6-Q10)

Answer: 041

1. With height h, those centers can be $(6,8,0),(0,8,h/2),(6,0,h/2)$. Two side vectors are $(-6,0,h/2),(0,-8,h/2)$.

2. Their cross product is $(4h,3h,48)$, so area $\frac{1}{2}\sqrt{25h^2+2304}=30$.

3. Thus $h=36/5>0$, and $m+n=41$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-I-Q10] | Part B (Q6-Q10)

Answer: 058

1. If the moving center sweeps angle ϕ around E, no-slip rolling causes rotation 5ϕ relative to the line of centers, plus ϕ from the turning line itself. Hence ground-frame rotation is 6ϕ, and one turn gives $\phi = 60^\circ$.

2. Take $E = (0, 0)$, initial moving center $(6, 0)$, and $A = (5, 0)$. The final center is $(3, 3\sqrt{3})$. A full rotation restores the marked point offset $(-1, 0)$, so $B = (2, 3\sqrt{3})$.

3. Thus the squared sine is $27/(4 + 27) = 27/31$, yielding 58.

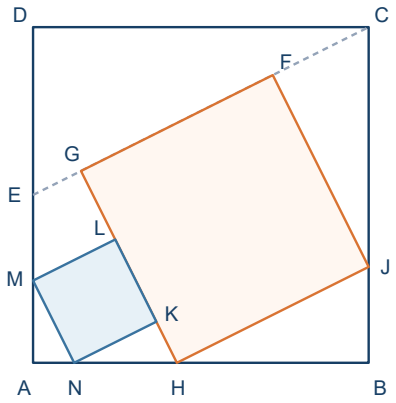
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Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-I-Q07] | Part B (Q6-Q10)

Answer: 539



1. Normalize the outer square to $A=(0,0), B=(1,0), C=(1,1), D=(0,1)$. Its line CE is $y=x/2+1/2$. Write $H=(h,0)$ and $J=(1,(1-h)/2)$ because $HJ \parallel CE$.

2. Rotating HJ by 90° toward CE gives $G=(h-(1-h)/2, 1-h)$. Requiring G on CE yields $h=3/7$. Consequently $[FGHJ]=(1-h)^2[1+(1/2)^2]=20/49$.

3. Line GH is $y=-2x+6/7$. Put $N=(u,0), M=(0,2u)$; the second square then has $K=(3u,u), L=(2u,3u)$. Its K incidence gives $u=6/49$, hence $[KLMN]=5u^2=180/2401$.

4. The area ratio is $(20/49)/(180/2401)=49/9$. Rescaling the inner area to 99 gives outer inner-square area $99\cdot 49/9=539$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2016-II-Q07] | Part B (Q6-Q10)

Answer: 840

1. Let side lengths be s, t, u, from outer to inner, and rotation angle be θ. Coordinate projections give $s = t(\cos\theta + \sin\theta)$ and $t = u(\cos\theta + \sin\theta)$. The same center follows from the equal extreme coordinates on the four supporting sides.

2. The areas form a geometric progression. If the middle area is T, the inner is $T^2/2016$, and T ranges continuously over $[1008, 2016)$ because $(\cos\theta + \sin\theta)^2$ ranges over $(1, 2]$.

3. For integral T, inner-area integrality requires $2016 \mid T^2$. Since $2016 = 2^5 \cdot 3^2 \cdot 7$, this is equivalent to $168 \mid T$. The allowed extreme multiples are 1008 and 1848, both attainable by rotation. Their difference is 840.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-II-Q10] | Part B (Q6-Q10)

Answer: 546

1. Use $A=(0,0), B=(84,0), C=(84,42), D=(0,42)$. Then $CM: y=x/4+21$ and $DN: y=42-3x/2$ intersect at $O=(12,24)$; $N=(28,0)$.

2. Shoelace gives $[BCON]=2184$, with target half1092. Since $[BCO]=1512>1092$, the boundary intersection P lies on CO. From base $BC=42$, $[BCP]=1092$ requires horizontal distance52 from P to BC.

3. Thus $P_x=32, P_y=32/4+21=29$. Finally $[CDP]=(84/2)(42-29)=546$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-I-Q08] | Part B (Q6-Q10)

Answer: 147

1. Turn successive edge directions by sixty degrees. Closing the vector chain gives $EF = 2, FA = 16$. Coordinates may be $A = (0, 0), B = (6, 0), C = (10, 4\sqrt{3}), D = (5, 9\sqrt{3}), E = (-7, 9\sqrt{3}), F = (-8, 8\sqrt{3})$.

2. The opposite-side strip widths are $9\sqrt{3}, 11\sqrt{3}, 7\sqrt{3}$. Thus any disk has diameter at most the narrowest width $7\sqrt{3}$.

3. To attain it, take center $(3/2, 11\sqrt{3}/2)$ and radius $7\sqrt{3}/2$. Its distances to CD, FA, DE equal the radius; those to AB, BC, EF are $11\sqrt{3}/2, 5\sqrt{3}, 6\sqrt{3}$, all larger. It lies in all six inward half-planes, hence the disk fits.

4. Therefore $d^2 = 147$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2018-II-Q09] | Part B (Q6-Q10)

Answer: 184

1. For each fan triangle JVW, its centroid is two-thirds of the way from J to the midpoint of VW. Hence the centroid polygon is a homothety of ratio $2/3$ of the polygon of midpoints of $AB, BC, \ldots, GH$.

2. Those midpoints are $(4,3), (27/2,0), (23,3), (27,23/2), (23,20), (27/2,23), (4,20)$. Split their heptagon by vertical lines at $x = 4, 27/2, 23$.

3. Two trapezoids of heights $19/2$ and parallel lengths seventeen, twenty-three contribute 380; the right triangle at the right contributes $17 \cdot 4/2 = 34$. Thus midpoint-polygon area is 414.

4. The target area is $(4/9)414 = 184$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-I-Q09] | Part B (Q6-Q10)

Answer: 567

1. The height is 18. Set $D=(0,0), C=(u+2a,0), A=(a,18), B=(u+a,18)$, where $u=AB$. Point-line distances give $18u/\sqrt{324+a^2}=15$ and $18u/\sqrt{324+(u+a)^2}=10$.

2. Squaring and subtracting yields $u+2a=9u/5$, hence $a=2u/5$. Then $324=32u^2/25$, so $u=45\sqrt{2}/4$ and $CD=81\sqrt{2}/4$.

3. $K=9(AB+CD)=567\sqrt{2}/2$, giving $\sqrt{2}K=567$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q08] | Part B (Q6-Q10)

Answer: 125

1. The parallel-side separation is $9 + 16 = 25$, so the inradius is $25/2$. Put the center at the origin and the parallel sides at $y = \pm 25/2$. Then $P = (\pm 12, 7/2)$.

2. Writing $\alpha = \angle BAD \in (0, 90^\circ)$, orient AB as $x \sin\alpha + y \cos\alpha = -25/2$. Its distance condition rules out $x = 12$ and gives $24 \sin\alpha - 7 \cos\alpha = 15$.

3. Together with the unit-circle identity, this yields $625s^2 - 720s + 176 = 0$, for $s = \sin\alpha$. The roots are $4/5, 44/125$; the latter would make $\cos\alpha < 0$, so $s = 4/5$.

4. The side is height divided by sine, $25/(4/5) = 125/4$, and the perimeter is 125.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-I-Q08] | Part B (Q6-Q10)

Answer: 077

1. The first locus is the circle centered at $(25, 20)$ with radius 5. The second is the perpendicular bisector of $(k+4, 0)$ and $(k, 3)$, namely $8x - 6y - 8k - 7 = 0$.

2. Unique intersection means tangency, so the center-to-line distance is $|73 - 8k|/10 = 5$. Thus $k = 23/8, 123/8$, whose sum is $73/4$; the requested sum is 77.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q10] | Part B (Q6-Q10)

Answer: 156

1. Take $B=(0,0), C=(14,0), A=(5,12)$. Equidistance gives circumcenter $O=(7,33/8)$. To turn direction $(3,-4)$ vertical, the rotation has $(\cos\phi,\sin\phi)=(4/5,-3/5)$ or its negative.

2. The opposite-side condition selects $(4/5,-3/5)$: the directed cross products for $A\prime$ and B relative to $B\prime C\prime$ are 168 and $-189/4$. This gives $A\prime=(81/8,93/8), B\prime=(-43/40,201/40), C\prime=(81/8,-27/8)$.

3. Apply the shoelace formula to the six vertices in the stated order. The area is $1557/10=155.7$, whose nearest integer is 156.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-I-Q13] | Part C (Q11-Q15)

Answer: 731

1. An optimal route can be shortened to travel on one road and then straight off-road: any off-road part that returns to a road is no faster than an appropriate road route. Six-minute distances are 5 on-road or $7/5$ off-road.

2. For the positive x-road, the endpoint disks have radius $(7/5)(1-d/5)$ at $(d,0)$. Their union has the straight tangent boundary from $(5,0)$ to the origin-centered radius-$7/5$ disk. Its upper line is $7x+24y=35$; swapping axes gives $24x+7y=35$.

3. The lines meet at $(35/31,35/31)$. Between the x-axis and the diagonal, the first line forms the outer reachable boundary; the other half is its reflection. Thus one quadrant is the quadrilateral $(0,0),(5,0),(35/31,35/31),(0,5)$, of area $175/31$.

4. The full area is $700/31$, giving 731.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2000-II-Q11] | Part C (Q11-Q15)

Answer: 131

1. Let $v = AD = (1, 7)$ and $w = BC$. Equal legs imply $|w|^2 = 50$, so w ranges over $(\pm 1, \pm 7), (\pm 7, \pm 1), (\pm 5, \pm 5)$. The vectors have the same component normal to the bases and opposite base components; hence $AB \parallel v - w$.

2. Exclude $w = v$ (parallelogram), $w = -v$ (zero height), and $(1, -7), (-1, 7)$ (vertical or horizontal bases). The remaining slopes are $-1, -4/3, 3/4, 1, -1/2, -3, 1/3, 2$.

3. Each is attainable with sufficiently long integer base vectors parallel to $v - w$. The absolute sum is $119/12$, giving 131.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2002-I-Q11] | Part C (Q11-Q15)

Answer: 230

1. Unfold successive reflections into neighboring cubes. Choose coordinates so the initial ray is $(12t, 7t, 5t)$ from the origin.

2. A folded vertex is reached when all three coordinates are multiples of 12. The first coordinate forces t integral, and then $7t$ and $5t$ require $12 \mid t$. The first positive value is $t = 12$.

3. The total distance is $12\sqrt{12^2 + 7^2 + 5^2} = 12\sqrt{218}$, giving 230.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-I-Q15] | Part C (Q11-Q15)

Answer: 289

1. Put $B = (0, 0)$ and its bisector on the x-axis. With $c = AB, a = BC$, write $A = (c\cos t, c\sin t)$ and $C = (a\cos t, -a\sin t)$. The bisector theorem gives $D = (2ac\cos t/(a + c), 0)$.

2. The line DF is vertical, giving $F_y = -2ac\sin t/(a + c)$. The midpoint is $M = ((a + c)\cos t/2, (c - a)\sin t/2)$; intersecting ray BM with that vertical gives $E_y = 2ac(c - a)\sin t/(a + c)^2$.

3. Therefore $DE/EF = (a - c)/(2c) = 147/720 = 49/240$. The requested sum is 289.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-II-Q11] | Part C (Q11-Q15)

Answer: 578

1. Use $A=(0,0), B=(25,0)$. From $AC=7, BC=24$, the x-coordinate of C is $(49+625-576)/50=49/25$. The midpoint has x-coordinate $25/2$.

2. The isosceles point D is vertically above M by $\sqrt{225-625/4}=5\sqrt{11}/2$. Thus take DM as base: the distance of C from its vertical line is $25/2-49/25=527/50$.

3. The area is $\frac{1}{2}(5\sqrt{11}/2)(527/50)=527\sqrt{11}/40$, giving 578.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2003-II-Q14] | Part C (Q11-Q15)

Answer: 051

1. Equal parallel opposite sides make the hexagon centrally symmetric, so opposite vertex heights sum to 10. Thus $D_y = 10, E_y = 8$. Reflect horizontally so $b > 0$. If $C_y = 4, F_y = 6$, the first two equal edge vectors are $(b, 2), (-b, 2)$, while the third is $(q, 6)$ with $|q| < b$. Their consecutive cross products are $4b > 0$ and $-6b - 2q < 0$, contradicting convexity. Hence $C_y = 6, F_y = 4$.

2. Reflect horizontally if needed to take $b > 0$. Put $AB = (b, 2), BC = (p, 4), CD = (q, 4)$. Equal lengths and noncollinearity give $p = -q$ and $q^2 = b^2 - 12$. The angle at A gives $bq + 8 = -(b^2 + 4)/2$.

3. Thus $q < 0$ and $b\sqrt{b^2 - 12} = (b^2 + 20)/2$, yielding $b^2 = 100/3$, $q = -8/\sqrt{3}$, $p = 8/\sqrt{3}$.

4. The parallelogram $ABDE$ has area $8b$, and the two remaining triangles each area $4p$. Total $8(b + p) = 48\sqrt{3}$, so the sum is 51.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2004-II-Q15] | Part C (Q11-Q15)

Answer: 593

1. Track strip length L, thickness H, current horizontal position p and bottom-up layer index b. Initially $(L, H, p, b) = (1024, 1, 942, 1)$. A left-half square keeps p, b; a right-half square changes to $p \leftarrow L + 1 - p$, $b \leftarrow 2H + 1 - b$. Then halve L and double H.

2. The successive (p, b) states are $(83, 2), (83, 2), (83, 2), (46, 15), (19, 18), (14, 47), (3, 82), (3, 82), (2, 431), (1, 594)$.

3. The final square is layer 594 from the bottom, so 593 squares lie below it.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-I-Q14] | Part C (Q11-Q15)

Answer: 936

1. The convex order forces A, C on opposite sides and B, D on opposite sides. If the side through B has slope m, its line is $mx - y + 9 - 10m = 0$; a perpendicular side through C has line $x + my - 8 = 0$.

2. Equal widths give $|2 - 14m| = |12m - 8|$, so $m = 5/13$ or -3. The first places A, C on opposite sides of the line through B, invalid for a supporting side. For $m = -3$ all four points lie on their required side segments.

3. The side length is $44/\sqrt{10}$, so $10K = 1936$ and the remainder is 936.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2005-II-Q12] | Part C (Q11-Q15)

Answer: 307

1. Put the side midpoint at horizontal coordinate zero and O height 450. Let signed coordinates of E, F be e, f, with $f - e = 400$. The angle tangent formula gives $1 = 450(f - e)/(450^2 + ef)$, so $ef = -22500$.

2. The roots with difference 400 are $e = -200 - 50\sqrt{7}$, $f = 200 - 50\sqrt{7}$; this orientation follows from $AE < BF$. Thus $BF = 450 - f = 250 + 50\sqrt{7}$.

3. The requested sum is 307.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2006-I-Q14] | Part C (Q11-Q15)

Answer: 183

1. Use top $T = (0, 0, 4)$ and feet $A = (3, 0, 0)$, $B = (-3/2, 3\sqrt{3}/2, 0)$, $C = (-3/2, -3\sqrt{3}/2, 0)$. Shortening TA by one gives $A\prime = (12/5, 0, 4/5)$.

2. The new supporting plane contains the vertical-y line BC. In the xz cross-section its equation through $(-3/2, 0), (12/5, 4/5)$ is $8x - 39z + 12 = 0$. The height is the distance from T to this plane, $144/\sqrt{1585}$.

3. $1585 = 5 \cdot 317$ is squarefree and $39 < \sqrt{1585} < 40$, so the requested floor is 183.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-I-Q14] | Part C (Q11-Q15)

Answer: 037

1. Scale the side to 2 and set $r = 1 + \sqrt{2}$, the apothem. Rotate coordinates so $M_1 = (0, r)$ and $M_3 = (-r, 0)$. An inward direction of R_3 is $(\cos\phi, \sin\phi)$; perpendicularity and the inward condition give direction $(\sin\phi, -\cos\phi)$ for R_1. The other rays are the successive quarter-turns.

2. Writing $B_1 = M_1 + a(\sin\phi, -\cos\phi) = M_3 + b(\cos\phi, \sin\phi)$ gives $a = r(\cos\phi - \sin\phi)$ and $b = r(\cos\phi + \sin\phi)$. Both are positive because the intersections lie on the rays. Rotation makes $M_3B_3 = a$, so $B_1B_3 = |a - b| = 2r|\sin\phi|$.

3. The required side length is 2, hence $\sin^2\phi = 1/r^2$. The vector M_3A_3 is vertical, so for $\beta = \angle A_3M_3B_1$ we have $\cos 2\beta = -\cos 2\phi = 2/r^2 - 1 = 5 - 4\sqrt{2} = 5 - \sqrt{32}$.

4. The choices $|\sin\phi| = 1/r < 1/\sqrt{2}$ give positive a, b and valid inward intersections, so this value is attained. Therefore $m + n = 37$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2011-II-Q13] | Part C (Q11-Q15)

Answer: 096

1. Set $A=(0,0), B=(12,0), C=(12,12), D=(0,12)$ and $P=(p,p)$, with $6<p<12$. Perpendicular-bisector equations give $O_1=(6,p-6), O_2=(6,p+6)$.

2. The vectors from P are $(6-p,-6),(6-p,6)$. The cosine condition is $[(p-6)^2-36]/[(p-6)^2+36]=-1/2$, giving $(p-6)^2=12$.

3. The stated inequality chooses $p=6+2\sqrt{3}$. Hence $AP=p\sqrt{2}=\sqrt{72}+\sqrt{24}$, and $a+b=96$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2012-I-Q13] | Part C (Q11-Q15)

Answer: 041

1. Let $OA = 3, OB = 4, OC = 5$. Rotate sixty degrees about A in the direction taking B to C, and call the image of O point P. Then $AP = OP = 3$ and $PC = 4$, so triangle OPC is a right 3-4-5 triangle at P.

2. The rays PA, PO make sixty degrees and PO, PC make ninety. Consequently $\angle APC$ is either thirty or one hundred fifty degrees. The cosine rule gives the complete possibilities $s^2 = 25 \pm 12\sqrt{3}$. Both occur by constructing the equilateral AOP on either side of OP and undoing the rotation to obtain B.

3. The larger area is $(\sqrt{3}/4)(25 + 12\sqrt{3}) = 9 + (25/4)\sqrt{3}$. Hence $a + b + c + d = 41$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2012-II-Q13] | Part C (Q11-Q15)

Answer: 677

1. Put $s^2 = 111, r^2 = 11$. The two rays AD_i are at signed angles $\pm\theta$ to AB. Rotating each by $\pm 60^\circ$ produces the four AE_k rays, all of length s.

2. Relative to ray AC at 60°, the four angle cosines sum to $2\cos\theta + \cos(120^\circ - \theta) + \cos(120^\circ + \theta) = \cos\theta$.

3. Thus the squared-distance sum is $2s^2(4 - \cos\theta)$. In triangle ABD_i, $r^2 = 2s^2(1 - \cos\theta)$, so this simplifies to $6s^2 + r^2 = 677$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-I-Q12] | Part C (Q11-Q15)

Answer: 021

1. The corner triangle BQC is equilateral, so take $Q=(0,0), C=(1,0), D=(2,0)$. The other vertices are $B=(1/2,\sqrt{3}/2), A=(1,\sqrt{3}), F=(2,\sqrt{3}), E=(5/2,\sqrt{3}/2)$.

2. The angle at R is forty-five degrees, so RP is a line $x+y=t$ supporting the hexagon. The greatest value of $x+y$ at a hexagon vertex occurs at F, hence $t=2+\sqrt{3}$; this also rules out touching only E.

3. Thus $QR=2+\sqrt{3}$. Intersecting with $QP: y=\sqrt{3}x$ gives height $(3+\sqrt{3})/2$. The area is $(9+5\sqrt{3})/4$.

4. The parameter sum is 21.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2013-II-Q13] | Part C (Q11-Q15)

Answer: 010

1. Set $B=(0,0), C=(t,0), A=(u,v)$, taking $v>0$. Equal sides give $u^2+v^2=2tu$. As $D=(t/4,0)$, the midpoint is $E=((u+t/4)/2, v/2)$.

2. The equation $BE^2=9$ becomes $(5/2)tu+t^2/16=36$. Subtracting BE^2 from CE^2 gives $tu=3t^2/4+2$.

3. Combining yields $t^2=16$, then $tu=14, u=7/2, v^2=63/4$. Thus area $tv/2=3\sqrt{7}$, and $m+n=10$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-I-Q13] | Part C (Q11-Q15)

Answer: 850

1. Use $B=(0,0), C=(s,0), D=(s,s), A=(0,s)$ and $E=(0,e), F=(f,0), G=(s,g), H=(h,s)$. The top two areas sum to half the square, so $e+g=s$. The left two sum to $2s^2/5$, so $f+h=4s/5$. Perpendicularity gives $h-f=e-g$.

2. Put $\delta=g-e$. Then $e=(s-\delta)/2, g=(s+\delta)/2, f=2s/5+\delta/2, h=2s/5-\delta/2$, and $s^2+\delta^2=1156$. Intersecting the lines gives $P=(s/2-s^3/11560, s/2-s^2\delta/11560)$.

3. The shoelace formula yields $[AEPH]=s^2/5-s^3\delta/231200$. Equate this with $269s^2/1360$ to obtain $s\delta=510$.

4. Thus s^2, δ^2 are roots of $t^2-1156t+260100=0$, namely $850, 306$. Since $|g-e|<s$, the square area is the larger value 850. The endpoint coordinates from these values lie on their required sides.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2014-II-Q11] | Part C (Q11-Q15)

Answer: 056

1. Take $D = (0, 0), R = (1/2, \sqrt{3}/2)$. The sine rule gives $E = ((1 + \sqrt{3})/2, 0)$ and $M = (1/4, \sqrt{3}/4)$.

2. Perpendicular slopes give the horizontal coordinate of C as $u = (7 - 3\sqrt{3})/11$, lying between D and E.

3. For $A = (t, 0)$, equate $(t - u)^2 = (t - 1/2)^2 + 3/4$. This gives $t = (1 - u^2)/(1 - 2u) = (9 + 4\sqrt{3})/11$, which lies beyond E.

4. Therefore $AE = (7 - 3\sqrt{3})/22 = (7 - \sqrt{27})/22$, giving 56.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2015-II-Q11] | Part C (Q11-Q15)

Answer: 023

1. Place B at the vector origin, and write vectors to A, C, O as a, c, o. Equal circumradii give $2o \cdot a = |a|^2 = 25$ and $2o \cdot c = |c|^2 = 16$.

2. The specified line has equation $o \cdot x = |o|^2$. If $P = ta, Q = sc$, then $25t = 16s = 2|o|^2$, so both parameters are positive and $t/s = 16/25$.

3. It follows that $BP/BQ = (5t)/(4s) = 4/5$. Thus $BP = (4/5)(9/2) = 18/5$, giving $m + n = 23$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2016-I-Q14] | Part C (Q11-Q15)

Answer: 574

1. The segment direction reduces to $(7, 3)$, repeated143 times, because $\gcd(1001, 429) = 143$. Its supporting line is $3x - 7y = 0$.

2. For square center (u, v), the linear form ranges within $|3|/10 + |7|/10 = 1$ of $3u - 7v$, so intersection requires $|3u - 7v| \leq 1$. On one primitive segment the centers are $(0, 0), (2, 1), (5, 2), (7, 3)$, with the middle two touched at corners.

3. A circle can be met only if $|3u - 7v|/\sqrt{58} \leq 1/10$. Since $\sqrt{58} < 10$, this forces $3u - 7v = 0$, so only the two primitive endpoints qualify. No centers beyond the full segment endpoints can intersect it, by the $1/10$ coordinate extents. Every center on the segment contributes a true circumference intersection, including the two full-segment endpoints, because the segment extends at least one radius into each corresponding disk.

4. There are144 lattice centers on the full segment and two additional square-only centers per primitive copy. Hence $m = 144 + 2 \cdot 143 = 430$, $n = 144$, and $m + n = 574$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-I-Q15] | Part C (Q11-Q15)

Answer: 145

1. Use outer vertices $(0,0),(5,0),(0,2\sqrt{3})$. If the equilateral side joining the legs has endpoints $(s\cos t,0),(0,s\sin t)$, its third vertex in the positive quadrant is $s(\cos t+\sqrt{3}\sin t,\sin t+\sqrt{3}\cos t)/2$. The other orientation has a negative coordinate and cannot lie in the triangle.

2. The hypotenuse equation is $2\sqrt{3}x+5y=10\sqrt{3}$. Substitution yields $s=20\sqrt{3}/(7\sqrt{3}\cos t+11\sin t)$. Cauchy bounds the denominator by $\sqrt{147+121}=2\sqrt{67}$.

3. Equality has $\cos t=7\sqrt{3}/(2\sqrt{67}),\sin t=11/(2\sqrt{67})$. The leg intercepts are $105/67,55\sqrt{3}/67$, within the side segments, and the third vertex is $(135/67,80\sqrt{3}/67)$ on the hypotenuse segment. Thus the bound is attained.

4. The minimum area is $(\sqrt{3}/4)(300/67)=75\sqrt{3}/67$, so $m+n+p=145$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2017-II-Q15] | Part C (Q11-Q15)

Answer: 682

1. Embed the tetrahedron at $A=(a,b,c), B=(a,-b,-c), C=(-a,b,-c), D=(-a,-b,c)$. The edge equations give $a^2+b^2=196$, $a^2+c^2=484$, $b^2+c^2=676$, so $(a^2,b^2,c^2)=(2,194,482)$. All are positive, and these coordinates realize every given edge.

2. Each vertex vector v has length $R=\sqrt{678}$, and the four vectors sum to0. For any position vector x, Cauchy gives $|v-x|\geq(v-x)\cdot(v/R)=R-v\cdot x/R$.

3. Sum this inequality over the four vertices to obtain $f(X)\geq 4R$. Equality is attained at $x=0$, so the minimum is $4\sqrt{678}$. As $678=2\cdot 3\cdot 113$ is squarefree, $m+n=682$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q11] | Part C (Q11-Q15)

Answer: 011

1. Take $A=(0,0), B=(7,0), C=(33/7, 24\sqrt{5}/7)$, using cosine rule. Tangency and passage through the third point give circle equations $x^2+y^2-7x+77y/(8\sqrt{5})=0$ and $x^2+y^2-189y/(8\sqrt{5})=0$.

2. Subtract to get the radical axis $y=(4\sqrt{5}/19)x$. Its positive unit direction is $(19/21, 4\sqrt{5}/21)$. Put $K=t$ times this vector in the second circle equation: $t^2=(189/(8\sqrt{5}))t(4\sqrt{5}/21)$.

3. The nonzero solution is $t=9/2$, hence $m+n=11$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2019-II-Q13] | Part C (Q11-Q15)

Answer: 504

1. Let side length be s, and u_j the outward unit normal toward midpoint of side A_jA_{j+1}. Signed triangle-area subtraction from the one-eighth sector gives region area $F_j = 1/8 - (s/2)P \cdot u_j$, valid even if P lies beyond the chord.

2. The given values imply $(s/2)P \cdot u_1 = -1/56$, $(s/2)P \cdot u_3 = 1/72$. Normals differ by forty-five-degree steps, so $u_6 = -(u_1 + u_3)/\sqrt{2}$.

3. Thus $(s/2)P \cdot u_6 = -(-1/56 + 1/72)/\sqrt{2} = \sqrt{2}/504$, giving $n = 504$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-I-Q15] | Part C (Q11-Q15)

Answer: 058

1. With circumcenter O as vector origin, $H = A + B + C$. The circle HBC is the reflection of ω across BC, so its center is $B + C = H - A$. Its radius toward H is therefore parallel to OA, and $XY \perp OA$.

2. Set $A = (0, R)$ and $H = (h, k)$ with chord XY horizontal. Since H is inside the circle and HX, HY are $2, 6$, the half-chord is 4 and $h^2 = 4$. Thus $R^2 - k^2 = 16$, while $HA^2 = 4 + (R - k)^2 = 9$. Hence $R - k = \sqrt{5}$ and $R = 21/(2\sqrt{5})$.

3. The midpoint M of BC is $(H - A)/2 = (h/2, -\sqrt{5}/2)$, giving $OM = 3/2$. Hence $BC = 2\sqrt{R^2 - 9/4} = 2\sqrt{99/5}$.

4. The distance from A to the chord line BC, whose normal is OM, is $|A \cdot M - |M|^2|/|M| = (21/4 + 9/4)/(3/2) = 5$. Thus the area is $5\sqrt{99/5} = 3\sqrt{55}$, and the answer is 58.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2020-II-Q12] | Part C (Q11-Q15)

Answer: 248

1. Use cell centers $(\text{column}, \text{row})$. The endpoints are $(200, 1)$ and (b, m), where $b = 2000 - (m-1)n$. Since m, n are odd, b is even. Their midpoint is the center of cell 1100; cell 1099 is immediately to its left.

2. The line reaches the left neighbor interior exactly when its absolute slope is below 1: $m - 1 < |b - 200| = |1800 - (m-1)n|$. Also $n \geq 201$ and $(m-1)n < 2000 \leq mn$, so $m = 3, 5, 7, 9$.

3. The odd-column ranges are respectively $667 \leq n \leq 999$, $401 \leq n \leq 499$, $287 \leq n \leq 333$, $223 \leq n \leq 249$. Their sizes are $167, 50, 24, 14$.

4. The slope test excludes respectively $\{899, 901\}, \{449, 451\}, \{299, 301\}, \{225\}$. The valid total is $165 + 48 + 22 + 13 = 248$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2021-II-Q12] | Part C (Q11-Q15)

Answer: 047

1. Let the diagonal vectors be $u = C - A$ and $v = D - B$. Expanding squared side lengths gives $BC^2 + DA^2 - AB^2 - CD^2 = 2u \cdot v = -21$. Hence $|u||v| \cos\theta = 21/2$.

2. The quadrilateral area is $|u \times v|/2$, so $|u||v| \sin\theta = 60$. Dividing gives $\tan\theta = 60/(21/2) = 40/7$, and the sum is 47.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-I-Q12] | Part C (Q11-Q15)

Answer: 075

1. Set $B=(0,0), C=(55,0), A=(55/2, 55\sqrt{3}/2)$. Then $D=(7,0), E=(40,15\sqrt{3}), F=(15/2, 15\sqrt{3}/2)$. The unit directions toward C, A, B from D, E, F are $u_3=(1,0), u_1=(-1/2, \sqrt{3}/2), u_2=(-1/2, -\sqrt{3}/2)$, with sum zero.

2. If d_1, d_2, d_3 are the perpendicular distances from P to CA, AB, BC, projection gives $(P-E)\cdot u_1 = d_1\cot\theta$ and its two analogues. Area decomposition gives $d_1+d_2+d_3 = 55\sqrt{3}/2$.

3. Add the projection equations. The terms involving P cancel, while $E\cdot u_1 = 5/2, F\cdot u_2 = -15, D\cdot u_3 = 7$. Therefore $(55\sqrt{3}/2)\cot\theta = -(5/2-15+7) = 11/2$.

4. Hence $\cot\theta = 1/(5\sqrt{3})$ and $\tan^2\theta = 75$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2023-II-Q12] | Part C (Q11-Q15)

Answer: 247

1. Take $B=(0,0), C=(14,0), A=(5,12), M=(7,0)$, so $AM=2\sqrt{37}$. The power relation $MA\,MP=49$ gives $P=(567/74,-147/37)$. Directions of BP, CP may be represented by $(27,-14),(-67,-42)$.

2. Write $Q=(5+2t,12-12t)$ with $0\le t\le 1$. Cross-product and dot-product formulas give the angle tangents $(394-296t)/(222t-33)$ at B, and $(1182-888t)/(370t+99)$ at C.

3. The angle at C is acute on this segment, so equality requires $222t-33>0$. The numerators have ratio 3, hence equality is equivalent to $370t+99=3(222t-33)$. It has the single valid root $t=99/148$.

4. Therefore $AQ=t\,AM=99/\sqrt{148}$. Since 99 and 148 are coprime, the answer is 247.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2025-I-Q12] | Part C (Q11-Q15)

Answer: 510

1. Factor the comparisons as $(x-y)(1+z)<0$ and $(y-z)(1+x)<0$. Of the four sign combinations, the bounded one is $-1<x<y<z$; the combination $x>y>z$ with $x,z<-1$ contradicts the plane, and the other two are unbounded.

2. The bounded-region closure has vertices $A=(-1,-1,77), B=(-1,38,38), C=(25,25,25)$, obtained by intersecting boundary lines pairwise.

3. Its area is half the norm of $(B-A)\times(C-A)$, namely $1014\sqrt{3}/2=507\sqrt{3}$. Thus $a+b=510$.

Source / 来源

Section 2.7 Coordinates, vectors & transformations

坐标、向量与变换

[AIME-2026-I-Q14] | Part C (Q11-Q15)

Answer: 681

1. Let side lengths be s_i in directions differing by 72°. Define $Q = \sum s_i^2 = 308$, $P_1 = \sum s_i s_{i+1}$, $P_2 = \sum s_i s_{i+2}$ cyclically. Diagonals are sums of successive side vectors, so $800 = 2Q + 2\cos 72^\circ P_1$.

2. Vector closure also gives $0 = Q + 2\cos 72^\circ P_1 + 2\cos 144^\circ P_2$. Thus $2P_1 = 184(\sqrt{5}+1)$ and $2P_2 = 492(\sqrt{5}-1)$.

3. The squared perimeter is $Q + 2P_1 + 2P_2 = 676\sqrt{5}$, hence $m + n = 681$.

Source / 来源